

Project Overview: Text-to-OksQuery Translation Module

1. PROJECT DESCRIPTION

The ATLAS Data Acquisition (DAQ) Configuration online service relies on the Object Kernel Support (OKS) framework to store and manage complex hardware and software topologies. To enable next-generation, AI-driven operations, a Model Context Protocol (MCP) server is being developed to allow Large Language Models (LLMs) to interact with this configuration data.

A critical challenge is querying the database. Pulling the entire configuration into an LLM's context window is technically unfeasible due to size constraints. Therefore, the MCP server must filter data on the backend using the native C++ oksQuery engine. Because LLMs do not natively understand the strict, proprietary syntax required by oksQuery, we require a specialized "translation" layer.

This project involves building a targeted prompt-engineering module that acts as a bridge, taking natural language search requests and accurately translating them into perfectly formatted oksQuery strings that the C++ backend can execute safely.

2. OBJECTIVE

The primary objective is to develop a standalone, highly accurate Natural Language to oksQuery translation module.

This module will serve as a middleware component for the larger MCP agent architecture. By handling the strict syntax formatting, this module allows the main AI Assistant to perform complex, filtered searches of the OKS database without needing to learn or memorize the underlying C++ query language, thereby reducing context waste and syntax hallucinations.

3. DELIVERABLES

At the conclusion of the project, the following deliverables are expected:

- 1. Translation Module (Python):** A standalone, importable Python script/module - a prompting module with schema retrieval (pulling in only the relevant part of the OKS schema, not all of it), few-shot examples, and a validate/repair loop.
- 2. Evaluation Dataset:** A curated set of common shifter questions paired with their "Ground Truth" (correctly formatted) oksQuery equivalents, stratified into difficulty levels.
- 3. Accuracy Benchmark Report:** A summary document or presentation detailing the translation accuracy of the module against the evaluation dataset, demonstrating its reliability for production integration. Accuracy should be measured by executing the generated query and comparing results to ground truth, not just comparing query text. Also report the valid-syntax rate. When reporting results, break down failures by an error taxonomy - by type (e.g. wrong syntax vs. wrong class/attribute selected), not just a single pass/fail percentage.

Ablation

Report accuracy in stages, showing what each part of the design contributes: plain prompt only, then + few-shot examples, then + schema retrieval, then + the validate/repair loop (full system).

4. RESOURCES THAT MAY BE HELPFUL

- **OKS Technical Paper:** [Object Kernel Support \(OKS\) framework overview](#)
- **OKS Repository Source:** [ATLAS TDAQ OKS GitLab Repository](#)